Robotic Technologies for the Objective Assessment of Spasticity: A Systematic Review

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Introduction

Spasticity is a motor disorder characterized by velocity-dependent increase in muscle tone and exaggerated stretch reflexes, affecting approximately 12 million people worldwide [1, 2]. As a major cause of impaired mobility and reduced functional independence, its accurate assessment is essential for diagnosis, treatment planning, and monitoring rehabilitation outcomes [3]. However, current clinical evaluation methods rely on subjective clinical scales that depend on examiner judgement and provide limited precision. To overcome these limitations, numerous technological solutions have been developed for the objective quantification of spasticity [4]. Nevertheless, their methodological foundations remain poorly described. Therefore, this systematic review provides a comprehensive overview of robotic devices for the objective assessment of spasticity by addressing the following research questions based on the PICO framework: (1) which technological systems currently exist for an objective quantification of spasticity? and (2) what methodological approaches have been integrated into technological systems to quantify spasticity?

Methods

A systematic search was conducted across five scientific databases following PRISMA guidelines. 1.106 articles were identified and independently analyzed by two reviewers. A total of 55 studies met the inclusion criteria and were analyzed in full-text. Data extraction focused on technological characteristics, including device type and sensor configuration, as well as methodological aspects such as signal processing strategies, mathematical models, and outcome measures used to quantify spasticity.

Results

The results of the literature review reveal that 48,7% of the technologies identified were exoskeletons, followed by end-effector devices that represent the 35,9%. Other technologies used to assess spasticity include isokinetic dynamometers and haptic devices.

Regarding the methodological approaches, 37,9% of the studies relied on simple feature extraction from force, electromyographic (EMG), or kinematic signals, whereas 53,4% applied system identification methods to estimate spasticity-related parameters (Figure 1). These system identification approaches generally involved three stages of analysis: (i) comparison across experimental conditions, (ii) implementation of biomechanical models, such as the second-order mass-spring-damper model, custom-developed models, or Hill-type muscle models, and (iii) parameter estimation through linear regression, machine learning algorithms, or optimization techniques. These procedures allow the quantification of spasticity.

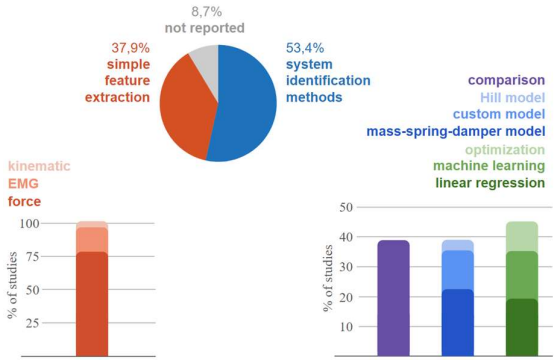


Figure 1: Methodological approaches separated into simple feature extraction (left), and system identification methods (right).

Conclusions

Exoskeletons and end-effector devices are currently the predominant robotic technologies for objective spasticity assessment. While simple feature extraction methods are still employed, system identification approaches are gaining prominence because they have the potential to accurately distinguish between neural and non-neural components of hyper-resistance. This review further aims to identify the biomedical parameters that most reliably detect spasticity and the methodological approaches that best capture the multidimensional nature of hyper-resistance. Preliminary findings highlight the need for more transparent reporting of assessment protocols to facilitate methodological standardization, validation, and clinical translation of emerging technologies.

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